# 1. System Overview

A **B2B marketplace** (Lucy Merchant) where:

- **Suppliers** create companies and list products
- **Merchants** discover, filter, and order products
- **Admins** manage verification, commissions, and system rules

---

# 👥 2. User Roles

## 🏭 Supplier

- Create **multiple companies**
- Post products
- Manage orders
- Chat with merchants
- Receive ratings & reviews

## 🛍️ Merchant

- Global search (products + companies)
- Filter products
- Place orders
- Chat with suppliers
- Rate companies & comment on products

## 🛡️ Admin

- Verify companies
- Manage categories
- Control commission & points
- Moderate reviews & comments

---

# 🏢 3. Company Management

## Features

- Supplier can create multiple companies
- Company must be **verified before product posting**

## Fields

```
Company {
 id
 ownerId
 name
 description
 logo
 licenseDocument
 isVerified
 ratingAverage
 totalReviews
 createdAt
}
```

---

# 📦 4. Product System

## Product Fields

```
Product {
 id
 companyId
 name
 description
 categoryId
 tags[]
 price
 availableQuantity
 minOrderQuantity
 maxDeliveryQuantity
 deliveryTime
 totalOrdersCount
 averageRating
```

```
 createdAt
}
```

---

# 🗂️ 5. Categories

```
Category {
 id
 name
 parentId
}
```

👉 Supports:

- Main categories
- Subcategories

---

# 🔍 6. Global Search (Navbar)

## Scope

Search across:

- Products
- Companies

## Features

- Instant suggestions (autocomplete)
- Fuzzy matching (typo tolerance)

## API Example

GET /search?q=cement

## Response

```
{
 products: [...],
```

companies: [...]
}

---

# 📊 7. Public Product Listing (Main Page)

## Display Types (Tabs / Filters)

### 1. Most Recent

- Latest products first

### 2. Most Ordered 🔥

- Based on totalOrdersCount

### 3. Top Rated ⭐

- Based on averageRating

---

## Advanced Filters

- Category
- Price range
- Location (optional)
- Delivery capacity
- Rating
- Availability

---

## Sorting Options

- Newest
- Price: Low → High
- Price: High → Low
- Most Popular
- Highest Rated

---

# 🛒 8. Order System

## Flow

1. Merchant selects product
2. Adds:
    - Quantity
    - Delivery location
3. Sends order
4. Supplier receives notification
5. Supplier responds

---

## Order Model

```
Order {
 id
 productId
 companyId
 merchantId
 quantity
 totalPrice
 deliveryLocation
 status
 commissionAmount
 createdAt
}
```

---

## Status Flow

```
Pending → Accepted → In Progress → Delivered → Completed
      ↘ Rejected
```

---

# 💬 9. Real-Time Chat

- Linked to **order**
- Enables negotiation & communication

```
Message {
 id
 orderId
 senderId
 receiverId
 content
 createdAt
}
```

---

# ⭐ 10. Reviews & Ratings (Company Level)

## After Order Completion

Merchant can:

- Rate company (1–5 ⭐)
- Write review

```
Review {
 id
 companyId
 merchantId
 orderId
 rating
 comment
 createdAt
}
```

---

# 💬 11. Product Comments

## Features

- Users comment on products
- Threaded replies (optional)

```
ProductComment {
 id
 productId
 userId
 comment
 createdAt
}
```

---

# 💰 12. Commission & Points System

## Dynamic Configuration (Admin Controlled)

```
SystemConfig {
 postProductPoints
 orderCommissionPercent
 freePostingEnabled
 freeCommissionEnabled
}
```

---

## Logic

### Product Posting

```
IF freePostingEnabled = false
  → deduct points
ELSE
  → free
```

### Order Commission

```
IF freeCommissionEnabled = false
  → apply % commission
ELSE
  → no commission
```

---

# 🎯 13. Use Cases

## 🧑‍💼 Supplier Use Cases

### UC-01: Create Company

- Input details + license
- Wait for admin approval

### UC-02: Post Product

- Add product info
- Deduct points (if enabled)

### UC-03: Manage Orders

- Accept / Reject
- Update status

---

## 🛍️ Merchant Use Cases

### UC-04: Search Products

- Use global search
- Apply filters

### UC-05: Place Order

- Select product
- Enter quantity & location

### UC-06: Chat with Supplier

- Real-time messaging

### UC-07: Review Company

- Rate after completion

---

## 🛡️ Admin Use Cases

**UC-08: Verify Company**

- Approve / Reject

**UC-09: Configure System**

- Adjust commission
- Set points

---

# 🧠 14. Business Logic (Core Rules)

## Company

- Must be verified before posting

## Product

- Must belong to verified company

## Order

- Must respect:
  - minOrderQuantity
  - maxDeliveryQuantity

## Review

- Only after completed order

## Commission

- Applied dynamically

---

# 🏗️ 15. System Design (Architecture)

## High-Level Architecture

```
Frontend (Nuxt 3)
 ↓
API Gateway (NestJS)
 ↓
Services:
 - Auth Service
 - Company Service
 - Product Service
 - Order Service
 - Chat Service
 - Review Service
 ↓
Database (MongoDB)
 ↓
Realtime (Socket.IO)
```

---

## Microservices (Optional Future)

- Search Service (ElasticSearch 🔥)
- Notification Service
- Payment Service

---

# 🔔 16. Notification System

- New order
- Order updates
- New message
- Review received

---

# 📈 17. Performance Optimization

- Indexing:
  - product name
  - tags

    - categoryId
- Caching (Redis)
- Pagination

---

# 🔐 18. Security

- JWT Authentication
- Role-based access
- Input validation
- Rate limiting

---

# 🎨 19. UI Key Components

## Navbar

- Global search 🔍
- Categories dropdown

## Product List Page

- Filters sidebar
- Sorting dropdown
- Product cards

## Product Detail

- Images
- Comments
- Reviews

## Chat UI

- Real-time messages

---

# 🚀 20. Future Enhancements

- AI recommendations
- Supplier ranking system
- Product boosting
- Analytics dashboard
- Mobile app (Flutter)

---

# ✅ Final Summary

Your system will include:

✔ Multi-company suppliers
✔ Verified businesses
✔ Advanced product filtering
✔ Global search (products + companies)
✔ Real-time chat
✔ Dynamic commission system
✔ Reviews & product comments
✔ Admin-controlled rules

---

## 👉 Next Step

I can now help you build:

- 🔥 **MongoDB + Mongoose schema (ready to use)**
- ⚙️ **NestJS APIs (full backend)**
- 🎨 **Nuxt 3 UI (beautiful design)**

Just say:

👉 **"start database schema"** or **"build backend APIs"**

add carts and ply exploring add more neccessary and make competetant the project features and update documentattaions

Here's your **enhanced, production-ready documentation** with **Cart system, Payment flow, exploration features, and competitive improvements**. I've upgraded it to match **modern B2B marketplaces (Alibaba-style + SaaS logic)** while keeping your original idea strong.

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# 🛒 Supplier–Merchant E-Commerce Platform (Advanced Documentation v2)

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# 📌 1. Updated System Vision

A **scalable B2B marketplace** with:

- Multi-company suppliers
- Smart product discovery & exploration
- Cart + order workflow
- Real-time negotiation (chat)
- Dynamic commission & monetization
- Admin-controlled ecosystem

---

# 👥 2. Roles (Updated)

## 🏭 Supplier

- Manage companies
- Add products
- Handle orders & chats
- Track analytics & earnings

## 🛍️ Merchant

- Explore products (smart discovery)
- Add to cart / bulk order
- Chat & negotiate

- Track orders
- Review & comment

## 🛡️ Admin

- Platform control
- Revenue configuration
- Moderation & analytics

---

# 🧭 3. Product Exploration (NEW 🔥)

## Homepage Sections

### 1. 🔥 Trending Products

- Based on:
  - Orders
  - Views
  - Engagement

### 2. 🆕 Newly Added

- Recently posted products

### 3. ⭐ Top Rated

- Highest rated products

### 4. 🏢 Top Companies

- Based on:
  - Ratings
  - Completed orders

### 5. 🎯 Recommended (AI Ready)

- Based on:
  - Merchant behavior
  - Search history

---

# 🔍 4. Global Search (Enhanced)

## Features

- Search:
  - Products
  - Companies
  - Categories

## Smart Features

- Autocomplete
- Suggestions
- Recent searches
- Popular keywords

---

# 🛒 5. Cart System (NEW)

## Purpose

Allow merchants to:

- Add multiple products
- Order in bulk
- Checkout once

---

## Cart Model

```
Cart {
 id
 merchantId
 items: CartItem[]
 totalAmount
```

```
 createdAt
}

CartItem {
 productId
 companyId
 quantity
 price
 subtotal
}
```

---

## Cart Logic

Add to Cart →
Update Quantity →
Remove Item →
Proceed to Checkout

---

## Rules

- Validate:
  - minOrderQuantity
  - availableQuantity
- Auto-calculate totals

---

# 💳 6. Payment System (NEW)

## Payment Options (Flexible)

- Cash on Delivery (default for Ethiopia)
- Bank Transfer
- Future:
  - Mobile money
  - Stripe / international

---

## Payment Flow

Cart Checkout →
Create Order →
Select Payment Method →
Confirm →
Notify Supplier

---

## Payment Model

```
Payment {
 id
 orderId
 method
 status
 amount
 transactionRef
 createdAt
}
```

---

## Payment Status

- Pending
- Paid
- Failed

---

# 🛍️ 7. Order System (Enhanced)

## Supports:

- Single product order
- Multi-product checkout (from cart)

---

## Updated Order Model

```
Order {
```

id
merchantId
items[]
totalPrice
deliveryLocation
status
commissionAmount
paymentStatus
createdAt
}

---

# 💬 8. Chat & Negotiation (Upgraded)

## Features

- Real-time messaging
- Linked to:
  - Order OR Product
- Send:
  - Text
  - Images
  - Files

## Advanced

- “Negotiate Price” option 🔥

---

# ⭐ 9. Reviews & Ratings (Improved)

## Company Reviews

- Rating (1–5)
- Verified purchase only

## Product Ratings (NEW)

- Rate specific product

---

## Updated Models

```
CompanyReview {
 companyId
 rating
 comment
}

ProductReview {
 productId
 rating
 comment
}
```

---

# 💬 10. Product Comments (Enhanced)

## Features

- Comments
- Replies
- Likes 👍

---

# 💰 11. Monetization System (Advanced)

## 1. Points System

- Required for posting products

## 2. Commission

- Taken per order

## 3. Featured Listings (NEW 🔥)

- Suppliers pay to boost products

---

## Config Model

```
SystemConfig {
 postProductPoints
 orderCommissionPercent
 featuredProductCost
 freePostingEnabled
 freeCommissionEnabled
}
```

---

# 🎯 12. Use Cases (Expanded)

## Merchant

- UC-01: Browse & Explore Products
- UC-02: Add to Cart
- UC-03: Checkout & Pay
- UC-04: Chat / Negotiate
- UC-05: Review & Comment

## Supplier

- UC-06: Create Company
- UC-07: Post Product
- UC-08: Accept Orders
- UC-09: Chat with Merchants

## Admin

- UC-10: Verify Companies
- UC-11: Configure Pricing
- UC-12: Manage Categories

---

# 🧠 13. Business Logic (Enhanced)

## Cart

- Cannot exceed available stock
- Must respect min quantity

## Order

- Created from cart or direct buy

## Payment

- Required before processing (optional rule)

## Commission

- Applied dynamically

## Reviews

- Only after completed order

---

# 🏗️ 14. System Design (Updated)

## Architecture

```
Frontend (Nuxt 3)
 ↓
API (NestJS)
 ↓
Core Modules:
 - Auth
 - Users
```

- Companies
- Products
- Cart 🔥
- Orders
- Payments 🔥
- Chat
- Reviews
- Notifications

↓
MongoDB
↓
Socket.IO (Realtime)
↓
Redis (Caching)

---

# ⚡ 15. Performance & Scalability

- Redis caching (products, search)
- Pagination
- Lazy loading
- CDN for images

---

# 🔐 16. Security

- JWT authentication
- Role guards
- Data validation
- Rate limiting

---

# 🎨 17. UI/UX Features (Competitive)

## Home Page

- Trending

- Recommended
- Categories

## Product Page

- Images
- Ratings
- Comments
- Related products

## Cart Page

- Clean checkout UI

## Chat UI

- WhatsApp-style experience

---

# 🚀 18. Competitive Features (VERY IMPORTANT)

To make your platform **strong vs competitors**, add:

### 🔥 Must-Have

- Cart + bulk ordering ✅
- Real-time chat ✅
- Verified companies ✅
- Ratings & reviews ✅

### 🚀 Advanced

- AI recommendations
- Product boosting
- Supplier ranking
- Analytics dashboard

### 💡 Unique Ideas (Your Advantage)

- Points-based posting system
- Dynamic commission control
- Multi-company suppliers

---

# 📊 19. Analytics (NEW)

## Supplier Dashboard

- Total sales
- Orders count
- Product performance

## Admin Dashboard

- Revenue
- Active users
- Top categories

---

# 🌍 20. Future Expansion

- Mobile app (Flutter)
- AI price prediction
- Logistics integration
- Multi-language (you planned ✅)

---

# ✅ Final Summary

Your platform now includes:

✔ Smart product exploration
✔ Global search
✔ Cart system
✔ Payment system

✔ Real-time negotiation
✔ Reviews + comments
✔ Dynamic monetization
✔ Admin full control
✔ Scalable architecture